COMPUTER ENGINEERING PROGRAM

DEGREE OFFERED

Bachelor of Engineering (Computer Engineering)
B.Eng. (Computer Engineering)

THE FIELD

Computer Engineering is defined as the discipline that embodies the science and technology of design, construction, implementation and maintenance of software and hardware components of modern computing systems and computer-controlled equipment. Computer engineering has traditionally been viewed as a combination of both computer science and electrical engineering. It has evolved over the past three decades as a separate, although intimately related, discipline. Computer engineering is solidly grounded in the theories and principles of computing, mathematics, science and engineering; it applies these theories and principles to solve technical problems through the design of computing hardware, software, networks and processes.

Technological advances and innovation continue to drive computer engineering. There is now a convergence of several established technologies (such as television, computer and networking technologies) resulting in widespread and ready access to information on an enormous scale. This has created many opportunities and challenges for computer engineers. This convergence of technologies and the associated innovation lie at the heart of economic development and the future of many organizations. The situation bodes well for a successful career in computer engineering.

CAREER OPPORTUNITIES

Increasingly, computer engineers are involved in the design of computer-based systems to address highly specialized and specific application needs. Computer engineers work in most industries, including the computer, aerospace, telecommunications, power production, manufacturing, defense and electronics industries. They design high-tech devices ranging from tiny microelectronic integrated-circuit chips to powerful systems that utilize those chips and efficient telecommunication systems that interconnect those systems. A wide array of complex technological systems, such as power generation and distribution systems and modern processing and manufacturing plants, rely on computer systems developed and designed by computer engineers.

CURRICULUM STRUCTURE

Courses	Credits
General Education Course	38
Core Courses	38
Major Required Courses	58
Major Elective Courses	8
I-Design Elective Courses	20
Free Elective Courses	8
TOTAL	170

COURSE LIST

Foundation Courses — Non-credit

ICMA 100 Foundation Mathematics 0 (4-0-0)
ICME 100 English Resource Skills 0 (4-0-0)

Note I: For B.Sc. and B.Eng. students, students whose Mathematics placement is below ICMA 106 Calculus I and/or ICMA 151 Statistics for Science I are required to take ICMA 100 Foundation Mathematics and pass the course with the grade of "S" before moving to ICMA 106 Calculus I and/or ICMA 151 Statistics for Science I.

Note II: Based on their achievement on the essay portion of the MUIC entrance exam, students whose English placement is below ICGC 101 Academic Writing and Research I will be placed into the "ERS Track". These students will be required to take ICME 100 English Resource Skills and pass the course with the grade of "S" before moving to ICGC 101 Academic Writing and Research I.

General Education Courses — 38 credits

English Communication — 12 - 16 credits

ICGC 101 Academic Writing and Research I 4 (4-0-8)
ICGC 102 Academic Writing and Research II 4 (4-0-8)
ICGC 103 Public Speaking 4 (4-0-8)
ICGC 201 Global Realities 4 (4-0-8)
ICGC 202 Literary Analysis 4 (4-0-8)
ICGC 203 Creative Writing 4 (4-0-8)
ICGC 204 Advanced Oral Communication 4 (4-0-8)
ICGC 206 Literature Into Film 4 (4-0-8)
ICGC 208 Language and Culture 4 (4-0-8)
ICGC 210 First and Second Language Acquisition 4 (4-0-8)
ICGC 211 Topics in Comparative Literature A: Poetry 4 (4-0-8)
ICGC 212 Topics in Comparative Literature B: The Short Story and the Novel 4 (4-0-8)

ICGC 213 Topics in Comparative Literature C: Drama	4 (4-0-8)
ICGC 214 Literary Non-fiction	4 (4-0-8)
ICGC 215 Writing for Research	4 (4-0-8)

Note I: Based on their achievement on the essay portion of the MUIC entrance exam, students will be placed into 3 following tracks: 'ERS Track', 'GC Track' and 'GC2 Track'.

- **'ERS Track'** – Students who are placed into 'ERS Track' will be required to take ICME 100 (a non- credit course) and complete 16 credits in English Communication: ICGC 101, ICGC 102, ICGC 103 in order, and finally, any 200+ level English courses.
- **'GC Track'** – Students who are placed into 'GC Track' will be required to complete 16 credits in English Communication: ICGC 101, ICGC 102, ICGC 103 in order, and finally, any 200+ level English courses.
- '**GC2 Track'** – Students who are placed into 'GC2 Track' will be required to complete 12 credits in English Communication: ICGC 102, ICGC 103 in order, and finally, any 200+ level English courses.

Note II: Students in 'ERS Track' and 'GC Track' must take ICGC 101, ICGC 102 and ICGC 103 without interruption beginning in their first trimester of enrollment.
Note III: Students in 'GC2 Track' must take ICGC 102 and ICGC 103 without interruption beginning in their first trimester of enrollment.

Life Appreciation	**4 credits**
ICGH 113 Moving Pictures: A History of Film	4 (4-0-8)
ICGH 117 Drawing as Creative Expression	4 (2-4-6)
ICGH 118 Photography Visualizing in the Digital Age	4 (2-4-6)
ICGH 119 Listen! Soundscapes, Well-Being and Musical Soul Searching	4 (4-0-8)
ICGN 105 Ecology, Ecosystems and Socio-Economics in Southeast Asia	4 (3-2-7)
ICGN 108 Essentials of Culinary Science for Food Business	4 (3-2-7)
ICGN 109 Food for Health	4 (4-0-8)
ICGN 110 Maker Workshop	4 (3-2-7)
ICGN 112 Stargazer	4 (3-2-7)
ICGN 113 Plants, People and Poisons	4 (4-0-8)
ICGN 115 Human Evolution, Diversity and Health	4 (4-0-8)
ICGN 120 Chemistry of Cosmetics and Dietary Supplements	4 (4-0-8)
ICGN 124 Climate Change and Human Society	4 (3-2-7)
ICGN 125 Games and Learning	2 (2-0-4)
ICGN 135 Personal Health and Well being	2 (2-0-4)
ICGP 101 American Flag Football	1 (0-2-1)
ICGP 102 Badminton	1 (0-2-1)
ICGP 103 Basketball	1 (0-2-1)
ICGP 104 Body Fitness	1 (0-2-1)
ICGP 105 Cycling	1 (0-2-1)
ICGP 106 Discover Dance	1 (0-2-1)

ICGP 107 Golf	1 (0-2-1)
ICGP 108 Mind and Body	1 (0-2-1)
ICGP 109 Selected Topics in Sports	1 (0-2-1)
ICGP 110 Self Defense (Striking)	1 (0-2-1)
ICGP 111 Self Defense (Grappling)	1 (0-2-1)
ICGP 112 Soccer	1 (0-2-1)
ICGP 113 Social Dance	1 (0-2-1)
ICGP 114 Swimming	1 (0-2-1)
ICGP 115 Tennis	1 (0-2-1)
ICGP 116 Volleyball	1 (0-2-1)
ICGP 117 Physical Fitness Plan	2 (2-0-4)
ICGS 102 Business Sustainability and the Global Climate Change	4 (4-0-8)
ICGS 111 Exploring Religions	4 (4-0-8)
ICGS 115 Sociology in the Modern World	4 (4-0-8)
ICGS 125 American History, Popular Media and Modern Life	4 (4-0-8)
ICGS 126 Introduction to Psychology	4 (4-0-8)
ICGS 127 Positive Psychology	4 (4-0-8)
ICGS 128 Global Gastronomy and Cuisines	4 (4-0-8)
ICGS 129 Tea Studies	2 (2-0-4)
ICGS 141 Coffee studies	2 (2-0-4)
ICGS 142 Contemporary Spirituality and Marketing	1 (1-0-2)
ICLL 100 Self Development	2 (2-0-4)

Global Citizenship	**4 credits**
ICGH 116 World Cinemas	4 (4-0-8)
ICGH 120 Thai and ASEAN Cinema	4 (4-0-8)
ICGH 121 The End of the World? Development and Environment	4 (4-0-8)
ICGH 122 Introduction to Asian Philosophy	4 (4-0-8)
ICGH 123 Faiths, Ecological Justice, and the Tropical Rainforests	2 (2-0-4)
ICGH 127 Rome: An Empire’s Rise and Fall	4 (4-0-8)
ICGL 101 Elementary German I	4 (4-0-8)
ICGL 102 Elementary German II	4 (4-0-8)
ICGL 103 Elementary German III	4 (4-0-8)
ICGL 111 Elementary Japanese I	4 (4-0-8)
ICGL 112 Elementary Japanese II	4 (4-0-8)
ICGL 113 Elementary Japanese III	4 (4-0-8)
ICGL 121 Elementary French I	4 (4-0-8)
ICGL 122 Elementary French II	4 (4-0-8)
ICGL 123 Elementary French III	4 (4-0-8)
ICGL 131 Elementary Chinese I	4 (4-0-8)
ICGL 132 Elementary Chinese II	4 (4-0-8)
ICGL 133 Elementary Chinese III	4 (4-0-8)
ICGL 141 Elementary Spanish I	4 (4-0-8)
ICGL 142 Elementary Spanish II	4 (4-0-8)

ICGL 143 Elementary Spanish III	4 (4-0-8)
ICGL 160 Introduction to Thai Language and Culture	4 (4-0-8)
ICGL 161 Elementary Thai I	4 (4-0-8)
ICGL 162 Elementary Thai II	4 (4-0-8)
ICGL 163 Elementary Thai III	4 (4-0-8)
ICGL 170 Diversities in Multilingual Societies	2 (2-0-4)
ICGL 201 Pre-intermediate German I	4 (4-0-8)
ICGL 202 Pre-intermediate German II	4 (4-0-8)
ICGL 203 Pre-intermediate German III	4 (4-0-8)
ICGL 211 Pre-intermediate Japanese I	4 (4-0-8)
ICGL 212 Pre-intermediate Japanese II	4 (4-0-8)
ICGL 213 Pre-intermediate Japanese III	4 (4-0-8)
ICGL 221 Pre-intermediate French I	4 (4-0-8)
ICGL 222 Pre-intermediate French II	4 (4-0-8)
ICGL 223 Pre-intermediate French III	4 (4-0-8)
ICGL 231 Pre-intermediate Chinese I	4 (4-0-8)
ICGL 232 Pre-intermediate Chinese II	4 (4-0-8)
ICGL 233 Pre-intermediate Chinese III	4 (4-0-8)
ICGL 241 Pre-intermediate Spanish I	4 (4-0-8)
ICGL 242 Pre-intermediate Spanish II	4 (4-0-8)
ICGL 243 Pre-intermediate Spanish III	4 (4-0-8)
ICGN 126 Plant Society	2 (2-0-4)
ICGS 106 Fashion and Society	4 (4-0-8)
ICGS 112 Geography of Human Activities	4 (4-0-8)
ICGS 123 Tourism Concepts and Practices	4 (4-0-8)
ICGS 130 Political Science	4 (4-0-8)
ICGS 131 Introduction to International Studies	4 (4-0-8)
ICGS 132 Career Preparation in a Globalized World	4 (4-0-8)
ICGS 133 Foundation of Mediterranean Cultures	4 (4-0-8)
ICGS 143 Introduction to Air Transport and Tourism	4 (4-0-8)

Critical Thinking	**4 credits**
ICGH 101 Biotechnology: from Science to Business	4 (4-0-8)
ICGH 102 Famous Arguments and Thought Experiments in Philosophy	4 (4-0-8)
ICGH 103 Logic, Analysis and Critical Thinking: Good and Bad Arguments	4 (4-0-8)
ICGH 105 Technology, Philosophy and Human Kind: Where Are We Now?!	4 (4-0-8)
ICGH 106 The Greeks: Crucible of Civilization	4 (4-0-8)
ICGH 107 Contemporary Art and Visual Culture	4 (4-0-8)
ICGH 109 Creative Thinking Through Art and Design	4 (2-4-6)
ICGH 110 Drawing as Visual Analysis	4 (2-4-6)
ICGH 115 Cinematic Languages and Its Application	4 (4-0-8)
ICGH 124 Life Drawing and Anatomy	4 (2-4-6)
ICGH 125 How Can We Know What Is Good? Moral Reasoning and Behavior	4 (4-0-8)
ICGH 126 Behavioral Ethics: Why Good People Do Bad Things	2 (2-0-4)

ICGN 107 The Chemistry of Everyday Life	4 (4-0-8)
ICGN 111 Physics for CEO	4 (4-0-8)
ICGN 123 The Earth's Dynamic Structure	4 (3-2-7)
ICGN 127 Practical Mathematics	2 (2-0-4)
ICGS 103 Economics in Modern Business	4 (4-0-8)
ICGS 105 Personal Financial Management	4 (4-0-8)
ICGS 113 Perspectives on the Thai Past	4 (4-0-8)
ICGS 134 Is Democracy Good?	4 (4-0-8)
ICGS 135 Entrepreneurial Accounting	4 (4-0-8)
ICGS 144 Price Discrimination: Why Do We Pay More Than Others?	1 (1-0-2)

Leadership	**4 credits**
ICGN 114 The Scientific Approach and Society	4 (4-0-8)
ICGN 128 Climate Emergency, Biodiversity Crisis, and Humanity at Risk	2 (1-2-3)
ICGS 104 Essentials of Entrepreneurship	4 (4-0-8)
ICGS 118 Skills in Dealing with People Across Cultures	4 (4-0-8)
ICGS 121 Abnormal Colleagues: How Do I Make This Work?	4 (4-0-8)
ICGS 136 Social and Health Issues in Thailand	4 (3-2-7)
ICGS 137 Witchcraft and Gender Representation	4 (4-0-8)
ICGS 138 Business Event Essentials	4 (4-0-8)
ICGS 139 Leadership and Change for a Global Society	4 (4-0-8)
ICGS 145 Service-Learning – Management of Community Service Project	4 (2-4-6)
ICGS 146 Service-Learning – Salaya Community Service Learning Project	4 (2-4-6)
ICGS 147 Women in Leadership	4 (4-0-8)
ICGS 148 Service Learning - Digital Campaign for Gender Issues	4 (2-4-6)
ICGS 149 Service Learning - Making Change	4 (2-4-6)
ICGS 150 Service Learning - Campaigning for a Cause	4 (4-0-8)
ICGS 151 Service Learning - The Art of Leadership in Practice	4 (2-4-6)
ICGS 152 Fantasy Literature’s Environmental Message	4 (4-0-8)
ICGS 154 Introduction to Crisis Management in Service Businesses	4 (4-0-8)
ICLL 101 Professional Development	2 (2-0-4)

Digital Literacy	**4 credits**
ICGH 111 Media Literacy: Skills for 21st Century Learning	4 (4-0-8)
ICGH 128 Internet Celebrity, Culture and the Media	4 (4-0-8)
ICGN 116 Understanding and Visualizing Data	4 (3-2-7)
ICGN 118 Everyday Connectivity	4 (4-0-8)
ICGN 119 Computer Essentials	4 (4-0-8)
ICGN 129 Programming for Problem Solving	4 (4-0-8)
ICGN 130 Cryptography: The Science of Making and Breaking Codes	2 (2-0-4)
ICGN 131 Digital Search Literacy	2 (2-0-4)
ICGN 132 Digital Security and Privacy	2 (2-0-4)
ICGN 133 E-Business: Technology and Digital Strategies	4 (4-0-8)
ICGN 134 Introduction to Artificial Intelligence	2 (2-0-4)
ICGS 140 Fake News, Censorship and the Politics of Truth	4 (4-0-8)

ICGS 153 Social Media Management for Service Sector	2 (2-0-4)
ICLL 102 Skills for a Digital World	2 (2-0-4)

General Education Elective Courses **2-6 Credits**

Students need to take any GE courses to fulfill their 38 credits requirement of General Education: 2 credits for students whose English Communication track are placed into 'ERS track' or 'GC Track' and 6 credits for students whose English Communication track are placed into 'GC2 Track'. Partial credits of GE course that exceed the GE requirements cannot be counted towards Free Electives.

Major Courses

Engineering Core Courses	**38 credits**
ICMA 106 Calculus I	4 (4-0-8)
ICMA 213 Calculus II	4 (4-0-8)
ICPY 101 Physics I	4 (4-0-8)
ICPY 102 Physics II	4 (4-0-8)
EGCI 113 Fundamental Computer Programming	3 (2-2-5)
EGCI 201 Discrete Mathematics	4 (4-0-8)
EGCI 211 Advanced Computer Programming	3 (2-2-5)
EGCI 230 Electric Circuit Analysis	4 (4-0-8)
EGCI 232 Engineering Electronics	4 (3-2-7)
EGCI 305 Statistics for Research in Computer Engineering	4 (4-0-8)

Major Required Courses	**58 credits**
ICMA 223 Linear Algebra A	2 (2-0-4)
EGCI 202 Engineering Maths for Signal and System	4 (4-0-8)
EGCI 213 Programming Paradigms	4 (4-0-8)
EGCI 221 Data Structures and Algorithms	4 (4-0-8)
EGCI 231 Digital Circuit Design	4 (4-0-8)
EGCI 233 Digital Circuit Design Lab	1 (0-2-1)
EGCI 252 System Programming	4 (4-0-8)
EGCI 319 Internship In Computer Engineering	2 (0-12-2)
EGCI 321 Database Systems	4 (4-0-8)
EGCI 330 Microprocessor and Interfacing	4 (4-0-8)
EGCI 332 Embedded Systems	4 (4-0-8)
EGCI 333 Computer Architecture	4 (4-0-8)
EGCI 340 Software Design	2 (2-0-4)
EGCI 341 Software Engineering	4 (4-0-8)
EGCI 351 Operating Systems	4 (4-0-8)
EGCI 371 Computer Networks and Information Security	4 (4-0-8)
EGCI 491 Computer Engineering Seminar	1 (0-2-1)
EGCI 492 Computer Engineering Project	2 (0-4-2)

Computer Engineering Elective Courses **8 Credits**

Note: To earn Specialization in each field, students must complete at least 12 credits from each specialization track.

Network and Security Elective Courses	
EGCI 432 Distributed Systems	4 (4-0-8)
EGCI 474 Internetworking Technologies I	4 (3-2-7)
EGCI 475 Internetworking Technologies II	4 (3-2-7)
EGCI 476 Cryptography and Computer Security	4 (4-0-8)
EGCI 477 Penetration Testing and Prevention	4 (4-0-8)
EGCI 478 Wireless Communication	4 (4-0-8)
System and Signal Elective Courses	
EGCI 431 Internet of Things	4 (4-0-8)
EGCI 451 Cloud Computing	4 (4-0-8)
EGCI 466 Big Data Processing	4 (4-0-8)
EGCI 467 Natural Language and Speech Processing	4 (4-0-8)
EGCI 486 Image Processing	4 (4-0-8)
EGCI 487 Computer Vision	4 (4-0-8)
Intelligent System and Robotic Sciences Specialization Courses	
EGCI 331 Introduction to IC Design	4 (4-0-8)
EGCI 406 Mechatronics	4 (3-2-7)
EGCI 407 Human-Robot Interaction (HRI)	4 (4-0-8)
EGCI 425 Data Mining	4 (4-0-8)
EGCI 461 Artificial Intelligence	4 (4-0-8)
EGCI 463 Pattern Recognition	4 (4-0-8)
Linguistic Software and Theory Elective Courses	
EGCI 301 Computer Graphics	4 (4-0-8)
EGCI 404 Theory of Computation	4 (4-0-8)
EGCI 427 Web Programming	4 (4-0-8)
EGCI 428 Mobile Device Programming	4 (4-0-8)
EGCI 429 Web Application Architecture	4 (4-0-8)
EGCI 494 Fundamental of Digital Forensics	4 (4-0-8)
Advanced Topics Courses	
EGCI 381 Introduction to Quantum Computing	4 (4-0-8)
EGCI 382 – 389 Special Topics in Computer Engineering (…)	4 (4-0-8)
EGCI 394 – 399 Special Topics in Computer Engineering (…)	4 (3-2-5)
Field Work Elective Course	
EGCI 493 Cooperative Education	8 (0-40-8)

EGCI 495 Regional Study Experience 1 (0-2-1)
EGCI 496 International Study Experience 2 (0-4-2)

I-Design Elective Courses 20 credits

The purpose of the I-Design electives is to promote multidisciplinary learning. Students are encouraged to explore courses offered by diverse disciplines across MUIC, Mahidol University, and partner institutions. The I- Design electives can be satisfied upon the completion of the following course categories:

1. Minor courses offered by any program in MUIC
2. Certificate courses offered by any program in MUIC
3. Any major courses offered by any program in MUIC
4. Any major courses offered in Mahidol University (including undergraduate and graduate level courses). Courses must be approved by the student's advisor and the Curriculum Administrative Committee.
5. Any major courses offered at partner institutions (who have MOU with Mahidol University and/or MUIC). Courses must be approved by the student's advisor and the Curriculum Administrative Committee.

Free Elective Courses 8 credits

Students can take any courses offered by MUIC and/or Mahidol University as free elective courses with the approval from the advisor. A course within the student's major that is too closely related or redundant to core/required/elective courses is discouraged and may be disapproved by an academic advisor.

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